25900326 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECSC201 - INTRODUCTION TO MOBILE AND WEB TECHNOLOGIES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define the term World Wide Web. [K] / [CO1]
2. What is a flexible grid layout? [K] / [CO2]
3. Define client-server model. [K] / [CO2]
4. Name two popular mobile development frameworks and briefly explain one key feature of each. [K] / [CO3]
5. If a business wants a single codebase that works across devices, why choose a PWA? [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Discuss the role of AI in today's mobile/web applications. [U] / [CO1]

7.	What is the process for running Node.js code, and what is one key requirement for its execution?	[U] / [CO2]
8.	What is React, and how does its component-based architecture and virtual DOM contribute to building modern web applications? Provide an example of a React component in use	[U] / [CO2]
9.	Explain the key stages of a mobile application's lifecycle	[A] / [CO3]
10.	Explain the importance of user experience (UX) in mobile applications and how it affects app success.	[U] / [CO3]
11.	Illustrate AES encryption and its use in mobile security.	[U] / [CO4]
12.	A developer needs to store authentication tokens securely on Android. Which secure storage mechanism should be used?	[U] / [CO4]
13.	List two security best practices for web application development, and briefly explain how each protects the application.	[U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14.	How do we create frames? Explain with suitable example.	[U] / [CO2]
15.	Provide a detailed analysis of the different types of mobile applications —native, hybrid, and web apps. Discuss their underlying technologies, development strategies, advantages, and disadvantages. Justify your selection for developing an e-commerce application for a startup	[E] / [CO3]
16.	Explain how AI-driven personalization improves user experience in mobile and web applications.	[U] / [CO4]

[15x2 = 30]